Maxime Newman Nereyabagabo

Burnaby, BC | (778) 814-0670 | maximenewman05@gmail.com | LinkedIn | GitHub | Website

Education

Simon Fraser University Jan. 2024 – Aug. 2028
Bachelor of Science, Computing Science *Burnaby, BC*

- Coursework: Machine Learning, Data Structures, Algorithms, Databases, Statistics, Networking, Systems Programming

Technical Skills

Languages: Python, Java, JavaScript, C, C++, C#, HTML/CSS, Bash, SQL
ML & AI: Scikit-learn, LangGraph, Pydantic, Pandas, NumPy, LangSmith
Frameworks & Libraries: Next.js, FastAPI, Django
Databases & Caching: PostgreSQL, Redis, Supabase
Cloud, DevOps & Automation: Docker, GitHub Actions, AWS, GCP, Azure, Git, CI/CD
Testing: Pytest, JUnit, Mockito, GoogleTest

Relevant Experience

SKC Engineering Ltd Jan. 2026 – May 2026
Data Engineering *Surrey, BC*

- Classified **15,639 SharePoint files** into **6 welding-process categories** with **70% requiring no manual review** by building an authenticated PowerShell pipeline via Azure App Registration, delivering a cleaned dataset to the ML team for fine-tuning
- Saved an estimated **$15,000+** in engineer time by building structured CSV logging with company-level and file-level outcomes, enabling full auditability and data quality tracking across parallel letter-batch migration runs
- Migrated **29,000+ rows** across **87 tables** from legacy Linode MySQL to managed DigitalOcean MySQL with **100% parity** by engineering a topological-sort foreign-key resolver, idempotent batched inserts, and symmetric pre/post-migration primary-key validation

SKC Engineering Ltd Sept. 2025 – Jan. 2026
AI Engineer *Surrey, BC*

- Built a full-stack AI-powered cost estimation platform with **GPT-4o mini** and **LangGraph**, using intent-based routing, deterministic tool execution, and structured state management to deliver consistent, auditable outputs
- Cut LLM inference cost per request by **20%** by stabilizing input size across long parameter-gathering sessions through rolling context windows, automatic summarization, and state-first prompting
- Improved model consistency across deployments with **50+ LangSmith eval cases**, using human-in-the-loop interrupts and a regression monitoring pipeline to detect behavioral drift
- Achieved **sub-300 ms API response times** for core authenticated endpoints by deploying to Fly.io using Docker multi-stage builds, GitHub Actions CI/CD, JWT-based access control, and idempotent APIs supporting a persistent multi-user workspace

Research Assistant Jan 2025 – Present
SFU CS ARCH Group *Burnaby, BC*

- Researched attention mechanisms and transformer architectures, focusing on how large-scale matrix multiplications dominate training and inference of LLMs.
- Authored 'read-along guide' on GPU-accelerated matrix multiplication kernel optimizations in CUDA, outlining concepts based on GPU multi-threading and execution models (thread blocks, warps, SM scheduling).
- Leveraged CUDA programming to design kernels in Linux environments to achieve 93.7% of cuBLAS using NVIDIA Nsight Compute (NCU) for analysis. Inspected PTX assembly output to further unravel optimizations, understanding IR passes done by the compiler itself.

Zebra Robotics Surrey Jan. 2025 – Jun. 2025
Robotics Instructor *Surrey, BC*

- Taught programming and robotics fundamentals using Python, Arduino, and microcontroller labs, enabling students to successfully program motors and sensors and complete functional robot prototypes
- Led student robotics projects from concept design through testing, guiding teams in mechanical iteration and debugging with programmable hardware, resulting in completed robots that met project goals and were showcased in a school exhibition

Projects

FraudLens | *Python, Pandas, Scikit-learn, NumPy* GitHub | SFU DSSS ML Hackathon

- Won **1st place** at the SFU DSSS ML Hackathon by building a multi-class fraud detection pipeline using Python, Pandas, and scikit-learn under tight time constraints
- Engineered 5+ predictive features using Pandas and NumPy, including **balance mismatch flags and error signals** to expose accounting inconsistencies, improving recall on rare minority fraud classes
- Trained a class-weighted **Random Forest** tuned against **Macro F1**, with a constraint-based data processing pipeline enforcing consistent types and flagging transaction impossibilities

Goblin's Keep | *Java, Maven, JUnit, Mockito* GitHub | Play Game

- Led backend development of a 2D tile-based escape game, architecting game logic and AI systems, establishing CI/CD pipelines with GitHub Actions, and coordinating code reviews across a team of developers
- Implemented dynamic goblin AI using a modified **A* pathfinding algorithm** with randomized patrol logic, enhancing difficulty balance and replayability
- Achieved **96% line coverage** and **92% branch coverage** with a **JUnit/Mockito test suite** ensuring correctness across all gameplay interactions

Other Experiences

Simon Fraser University Jan. 2025 – Apr. 2025
Calculus Teaching Assistant *Burnaby, BC*

- Led weekly peer-led sessions for a dedicated group of students, fostering collaborative learning and long-term academic growth
- Explained key concepts through interactive discussions and guided problem-solving sessions
- Reinforced personal mastery of calculus through teaching, feedback, and real-time troubleshooting

Simon Fraser University Aug. 2024 – Sept. 2024
Hive Leader *Burnaby, BC*

- Organized and led events to welcome and engage new students
- Collaborated with other Hive Leaders to plan and execute group activities
- Supported and mentored HIVE volunteers aspiring to become Hive Leaders
- Led orientation sessions, ensuring new students had a smooth transition

SFU Recreation Apr. 2024 – Aug. 2024
Recreation Assistant *Burnaby, BC*

- Front Desk Reception: Managed check-ins, answered inquiries, and kept daily operations running smoothly
- Customer Service: Built relationships with clients, addressed concerns, and made sure every member felt welcomed
- Fitness Class Assistant: Supported instructors with class setup and equipment while keeping participants motivated and sessions running safely
- Intramural Soccer League Assistant: Helped organize and run a soccer league, handling scheduling, officiating, and ensuring games ran smoothly